
MULTIMODAL VISION-LANGUAGE VARIATIONAL AUTOENCODER AS A COMPUTATIONAL MODEL OF THE HUMAN VISUAL SYSTEM

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ABSTRACT

Visual perception involves generative processing of sensory inputs. Variational Autoencoders (VAEs) have been used to model this process, but they often struggle to capture high-level semantic features and show weak representational alignment with higher visual areas. Because Large Language Models (LLMs) represent semantic information well, integrating them into VAEs may help address this misalignment. In this exploratory study, we aimed to improve VAE's representational alignment with the human visual system through the LLM guidance. We introduced a VAE whose latent variables are constrained by LLM embeddings of image captions. We evaluated the model using functional magnetic resonance imaging (fMRI) data from natural scene viewing. Our analysis showed that the LLM guidance improved the alignment of VAE latent variables with higher visual areas, but did not improve the VAE's overall representational alignment with them.

1 Introduction

A key aspect of visual perception is its generative nature. For example, the brain completes missing information based on available sensory inputs, such as filling-in at the blind spot [Pessoa et al., 1998, Komatsu, 2006]. It can also produce visual experiences in the absence of sensory inputs, as seen in mental imagery and dreaming [Dijkstra et al., 2019]. Furthermore, several theoretical frameworks argue that the brain constantly generates predictions of incoming sensory signals [Neisser, 1967, Rao and Ballard, 1999, Friston, 2003, Bastos et al., 2012, Aizenbud et al., 2025]. Generative mechanisms may also support visual tasks by providing teacher signals for self-supervised learning [Hinton et al., 1995, Dayan et al., 1995]. Taken together, modeling generative computation in the visual system is essential for understanding visual perception.

Artificial Neural Networks (ANNs) provide a powerful framework for investigating computation underlying brain functions [Kietzmann et al., 2019, Richards et al., 2019, Doerig et al., 2023]. In particular, Convolutional Neural Networks (CNNs) have been used to investigate the primate visual system [Kriegeskorte, 2015, Serre, 2019, Lindsay, 2021]. For example, optimizing CNNs for object classification aligns their internal representations with those in the visual system, suggesting that such optimization may shape neural coding [Yamins et al., 2014, Yamins and DiCarlo, 2016]. However, standard CNN models overlook the generative nature of visual perception [Serre, 2019, Linsley et al., 2025].

Variational Autoencoders (VAEs) have been used to model generative computation in the visual system [Han et al., 2019, Golan et al., 2020, Higgins et al., 2021, Chang et al., 2021]. VAEs are latent variable models that approximate the posterior distribution of latent variables given input image data [Kingma and Welling, 2013, Rezende et al., 2014]. Unlike Autoencoders (AEs), which encode inputs as deterministic latent variables, VAEs encode inputs as distributions, and sampling from them enables slightly different reconstructed images for the same input (Figure 1). While these properties make VAEs appealing for modeling generative computation, generative models like VAEs often struggle to learn high-level semantic representations [Nalisnick et al., 2019]. More crucially, Han et al. [2019] showed that the VAE’s representations aligned with human higher visual areas more weakly than CNN representations did. VAEs alone may be insufficient to model higher visual areas.

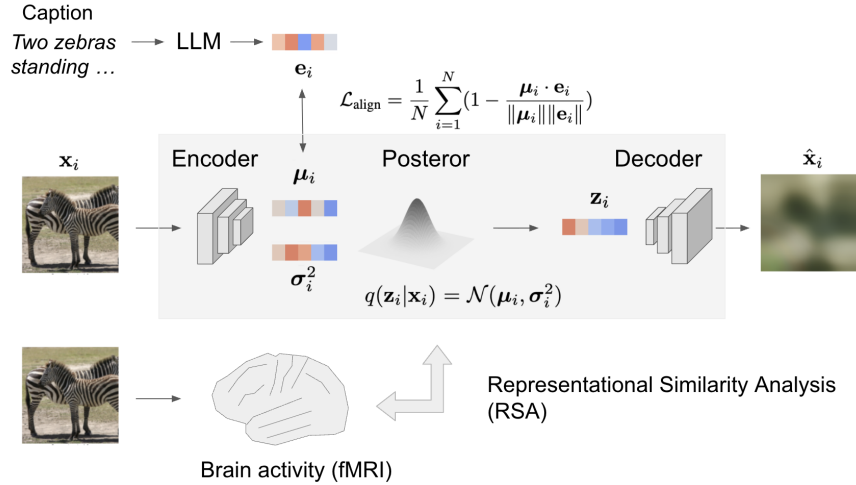


Figure 1. Integration of Large Language Model (LLM) embeddings into a Variational Autoencoder (VAE) for improving its representational alignment with the human visual system. The VAE encoder maps an image input x_i to a Gaussian posterior over latent variables $q(z_i|x_i) = \mathcal{N}(\mu_i, \sigma_i^2)$. A latent variable z_i drawn from the posterior is processed by the decoder to reconstruct the input image $\hat{x}_i$. We introduced an LLM alignment loss $\mathcal{L}_{align}$ to guide the posterior mean μ_i toward LLM embeddings e_i of the corresponding image caption. We evaluated the representational alignment with the human visual system via representational similarity analysis (RSA). See Methods for details.

Leveraging linguistic information may improve the VAE’s representational alignment with higher visual areas. Language and perception are tightly connected, as supported by behavioral and neural data [Kay and Kempton, 1983, Regier and Kay, 2009, Thierry, 2016, Mahon and Kemmerer, 2020, Lupyan et al., 2020]. Language has also been considered a top-down signal for the visual system [Lupyan, 2012, Lupyan and Clark, 2015, Lupyan et al., 2020]. Empirically, multimodal vision-language models predict brain activity better than unimodal vision models [Wang et al., 2023, Tang et al., 2023].

To integrate linguistic information into VAEs, we proposed to constrain the VAE’s latent variables by Large Language Model (LLM) embeddings (Figure 1). LLM embeddings capture rich semantic information from input text. These embeddings align with neural representations in human language areas [Schrimpf et al., 2021, Caucheteux and King, 2022, Goldstein et al., 2022] as well as higher visual areas [Doerig et al., 2025]. We leveraged these properties by training a VAE on images while directing its latent posterior means toward LLM embeddings of the corresponding image captions. This design also allows text-to-image generation through latent variables, which parallels mental imagery.

In this exploratory study, we investigated an LLM-guided VAE as a model of the human visual system. By directing the VAE’s latent posterior means toward LLM embeddings, we aimed to improve its representational alignment with higher visual areas. We evaluated the alignment using functional magnetic resonance imaging (fMRI) data collected during natural scene viewing. We found that the LLM guidance improved the alignment of VAE latent variables with higher visual areas, but not the VAE’s overall alignment with them.

2 Method

We briefly describe our models, with detailed information in Supplementary Material S1. We then describe our method for evaluating their representational alignment with the human visual system.

2.1 Variational Autoencoder (VAE) constrained by Large Language Model (LLM) embeddings

We implemented a convolutional Variational Autoencoder (VAE). We designed an encoder with ten convolutional blocks based on the vNet [Mehrer et al., 2021, Doerig et al., 2025], modeling the increase in receptive field size along the human visual system. We roughly mirrored the encoder architecture for the decoder. The encoder processes an image input $\mathbf{x}_i$ to estimate the mean $\boldsymbol{\mu}_i$ and variance $\boldsymbol{\sigma}_i^2$ of a Gaussian posterior over latent variables, $q(\mathbf{z}_i|\mathbf{x}_i) = \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\sigma}_i^2)$. A latent variable $\mathbf{z}_i$ is randomly sampled from the posterior through the reparameterization trick [Kingma and Welling, 2013], and the decoder reconstructs the input image $\hat{\mathbf{x}}_i$ from $\mathbf{z}_i$. See Supplementary Section S1.1 for the full architecture.

To integrate linguistic information into the VAE, we introduced an LLM alignment loss $\mathcal{L}_{\text{align}}$. We define $\mathcal{L}_{\text{align}}$ as the cosine distance between the latent posterior mean $\boldsymbol{\mu}_i$ and the LLM embedding $\mathbf{e}_i$ of the corresponding image caption, averaged across N inputs:

$$\mathcal{L}_{\text{align}} = \frac{1}{N} \sum_{i=1}^N \left(1 - \frac{\boldsymbol{\mu}_i \cdot \mathbf{e}_i}{\|\boldsymbol{\mu}_i\| \|\mathbf{e}_i\|}\right) \quad (1)$$

Minimizing the loss encourages directional alignment between $\boldsymbol{\mu}_i$ and $\mathbf{e}_i$. We combined it with the standard β -VAE loss [Higgins et al., 2016] to define a total loss $\mathcal{L}$:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta D_{\text{KL}}(q(\mathbf{z}|\mathbf{x})||p(\mathbf{z})) + \lambda \mathcal{L}_{\text{align}} \quad (2)$$

where $\mathcal{L}_{\text{recon}}$ is the $L2$ image reconstruction loss, β controls the Kullback-Leibler divergence D_{KL} between the approximated posterior $q(\mathbf{z}|\mathbf{x})$ and the standard Gaussian prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$, and λ controls the LLM alignment loss. $\mathcal{L}_{\text{recon}}$ is defined as follows.

$$\mathcal{L}_{\text{recon}} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 \quad (3)$$

We trained the model on the Microsoft Common Objects in Context (COCO) dataset [Lin et al., 2014], which provides five captions per natural scene image. We fed image captions into MPNet [Song et al., 2020] to obtain LLM embeddings, and averaged them per image for the loss computation. Following Doerig et al. [2025], we split the dataset to avoid overlap with a downstream evaluation dataset, yielding 48,236 pairs of images and corresponding LLM embeddings for training, 2,051 for validation, and 73,000 for testing. Supplementary Section S1.2 provides further details on the dataset.

To promote LLM alignment, we took a three-stage training procedure. In the first stage, we pre-trained the encoder to predict LLM embeddings. In the second stage, we kept the encoder frozen and pre-trained the decoder to reconstruct images from the predicted LLM embeddings. Finally, we initialized the full VAE with these weights and trained it using the reparameterization trick. We selected model weights at each stage based on validation performance. See Supplementary Section S1.3 for the detailed training procedure.

As reference models, we trained a standard VAE, an AE, and an LLM-guided AE. The standard VAE shared the architecture with the LLM-guided VAE, but was optimized without the LLM alignment loss $\mathcal{L}_{\text{align}}$ (Eq. 1). AEs had the same architecture but without the reparameterization trick. The standard AE was optimized with the $L2$ image

reconstruction loss $\mathcal{L}_{\text{recon}}$ (Eq. 3), while the LLM-guided AE was optimized with $\mathcal{L}_{\text{recon}}$ as well as $\mathcal{L}_{\text{align}}$ between AE latent variables and LLM embeddings. See Supplementary Section S1.4 for more details.

2.2 Evaluating representational alignment through Representation Similarity Analysis (RSA)

We evaluated representational alignment between the models and the human visual system using functional magnetic resonance imaging (fMRI) responses from the Natural Scenes Dataset (NSD) [Allen et al., 2022]. The dataset provides 7T fMRI data recorded from eight participants during a visual recognition task on 73,000 COCO images. The NSD includes estimates of fMRI response amplitudes to each stimulus as single-trial beta coefficients from General Linear Model (GLM) analysis at 1.8-mm isotropic resolution (betas_fithrf_GLMdenoise_RR). We included data for 515 images that all participants saw three times, and averaged beta values across three presentations for further analysis. We focused on the visual “streams” regions of interest (ROIs): early, midparietal, parietal, midlateral, lateral, midventral, and ventral areas.

To evaluate representational alignment, we employed a fitted Representational Similarity Analysis (RSA) approach [Kriegeskorte et al., 2008, Kaniuth and Hebart, 2022]. We first computed a fMRI representational dissimilarity matrix (RDM) for each ROI and participant. We defined each entry in the RDM as one minus Pearson’s correlation between voxel-wise beta coefficient patterns of two images. Similarly, we obtained an RDM for each model layer, defining each entry as one minus Pearson’s correlation between unit activation patterns of two images. We then fit an ordinary least squares linear regression model, using the model RDMs as predictors, to predict the fMRI RDM. As an RSA alignment score, we used the Spearman correlation between the predicted and actual fMRI RDMs. We performed this analysis per ROI and participant with five-fold cross-validation by splitting the 515 images into five folds of 103 images each. We averaged the resulting scores across five held-out folds. As a baseline, we estimated the noise ceiling as the Spearman correlation between each participant’s RDM and the average RDM of the remaining participants. We computed the correlation using data corresponding to each fold in the cross-validation, and averaged them to obtain the noise ceiling for each participant. We reported RSA alignment scores normalized by the participant-wise noise ceiling.

In the fitted RSA approach, we considered four conditions: encoder, latent variables, decoder, and all combined. Because the encoder and decoder consist of multiple layers, we included the last layer of each convolutional block for this analysis (see Supplementary Table 1). Out of the ten convolutional blocks in the decoder, we excluded the last block as it produces reconstructed images rather than internal representations. To match the number of blocks, we excluded the first encoder block, given the mirrored architecture. This resulted in nine RDMs for each of the encoder and the decoder conditions, and 19 RDMs for the all-combined condition.

We assessed whether LLM-guided models achieved higher RSA scores than corresponding standard ones. We performed one-tailed paired t -tests against zero across eight participants and applied the Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995] to control the false discovery rate (FDR) at 0.05 across 56 comparisons (7 ROIs $\times$ 4 layer conditions $\times$ 2 model pairs: VAE and AE). This was an exploratory analysis without a power analysis, and the statistical tests reported below should be interpreted descriptively.

3 Result

We trained standard and LLM-guided variants of VAE and AE (training history in Supplementary Figures 1–6), and assessed the effect of LLM alignment loss $\mathcal{L}_{\text{align}}$ (Eq. 1) by comparing standard and LLM-guided models per architecture. For the VAEs, the alignment loss made the posterior mean μ_i similar to the corresponding LLM embedding e_i for the test dataset (Figure 2a). However, the alignment loss reduced image reconstruction quality (Figures 2b and 2c). For the AEs, we observed the same pattern of increased LLM alignment and reduced image reconstruction quality. This suggests a general trade-off between LLM alignment and image reconstruction quality.

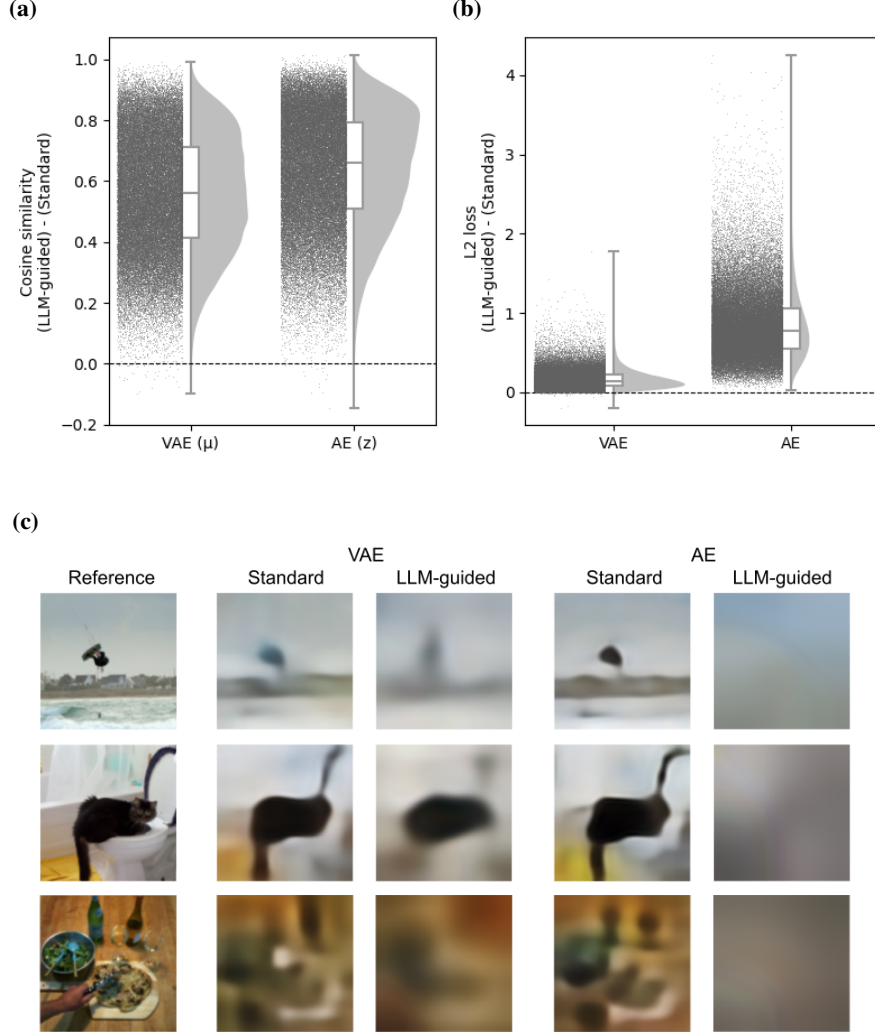


Figure 2. Our LLM alignment loss improved model alignment with LLM embeddings but reduced image reconstruction quality. **(a)** Difference in LLM alignment between LLM-guided and standard models. We computed cosine similarity between LLM embedding e_i and the posterior mean μ_i (VAE) or latent variables z_i (AE) for each of the 73,000 test image–embedding pairs. The difference is shown as a raincloud plot. **(b)** Difference in L_2 image reconstruction loss for the test dataset. **(c)** Examples of reconstructed images.

We evaluated representational alignment between the models and the human visual system through a fitted RSA approach. When considering all model components together (the all-combined condition), the LLM-guided VAE did not show higher RSA alignment scores than the standard VAE across visual areas (Figure 3a). In contrast, the LLM-guided AE showed higher RSA scores than the standard AE across visual areas, particularly in higher visual areas. We also evaluated representational alignment for each model component: the encoder, latent variables, and decoder (Figure 3b). For VAE latent variables (light gray bars), the LLM guidance improved RSA scores within higher visual areas, but this improvement did not change the VAE’s overall alignment as shown in Figure 3a. Unlike the VAE, the LLM guidance improved the AE’s representational alignment for the latent variables and decoder (light gray and dark gray bars). Table 1 shows results for an exploratory statistical analysis. Together, these results indicate that the LLM guidance has greater benefits for representational alignment with visual areas in the AE than in the VAE.

(a)

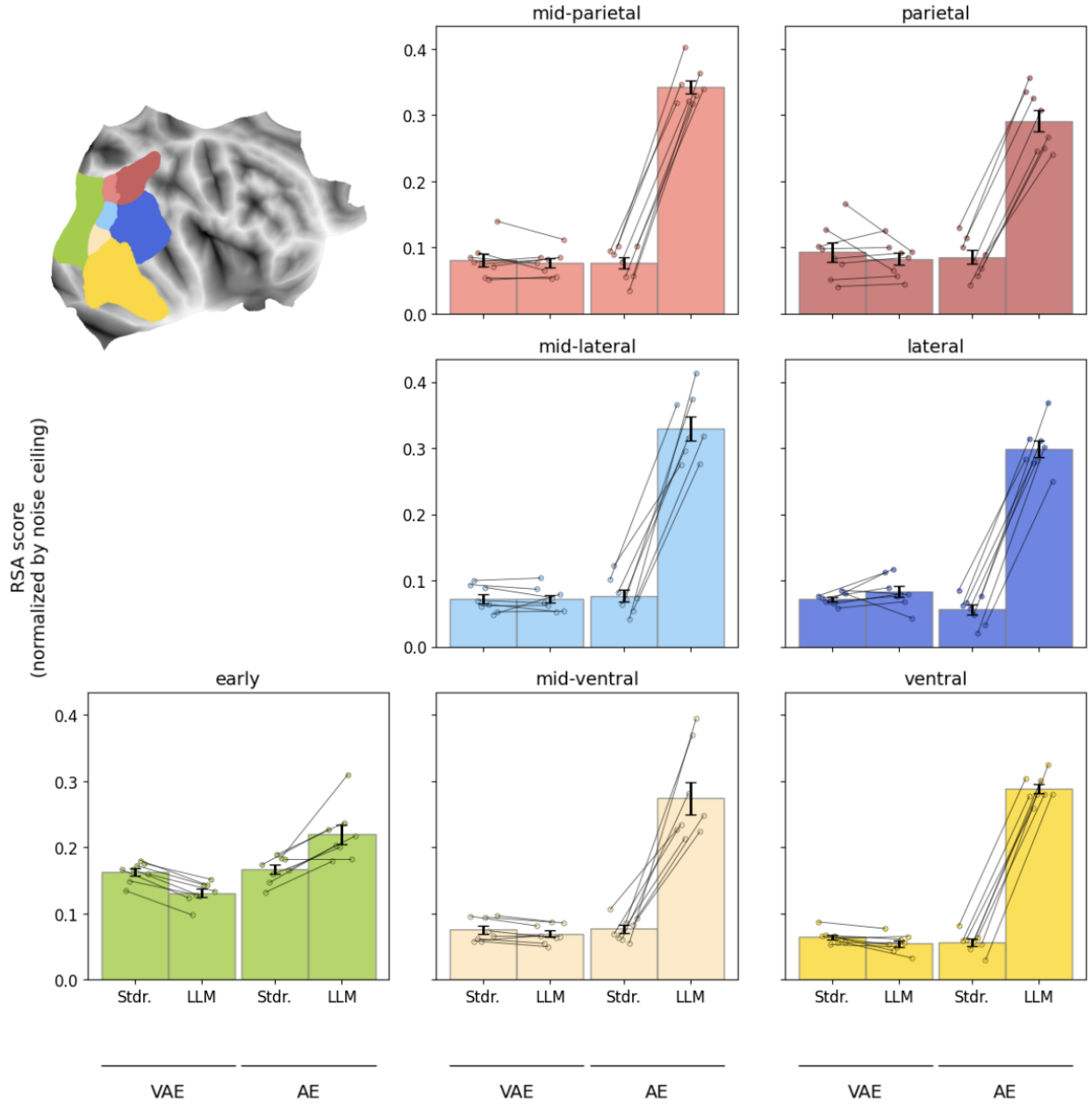


Figure 3. Representational alignment between our models and the human visual system, measured through a fitted Representational Similarity Analysis (RSA). For each region of interest (ROI), we predicted an fMRI Representational Dissimilarity Matrix (RDM) from model layer RDMs using an ordinary least squares linear regression model. We measured RSA scores as the Spearman correlation between the predicted and actual fMRI RDMs, normalized by the participant-wise noise ceiling. (a) RSA scores using model RDMs from all model components (all-combined) for each of the standard and LLM-guided VAE and AE models. Dots with connecting lines indicate individual participant data. Error bars show the standard error of the mean across eight participants.

(b)

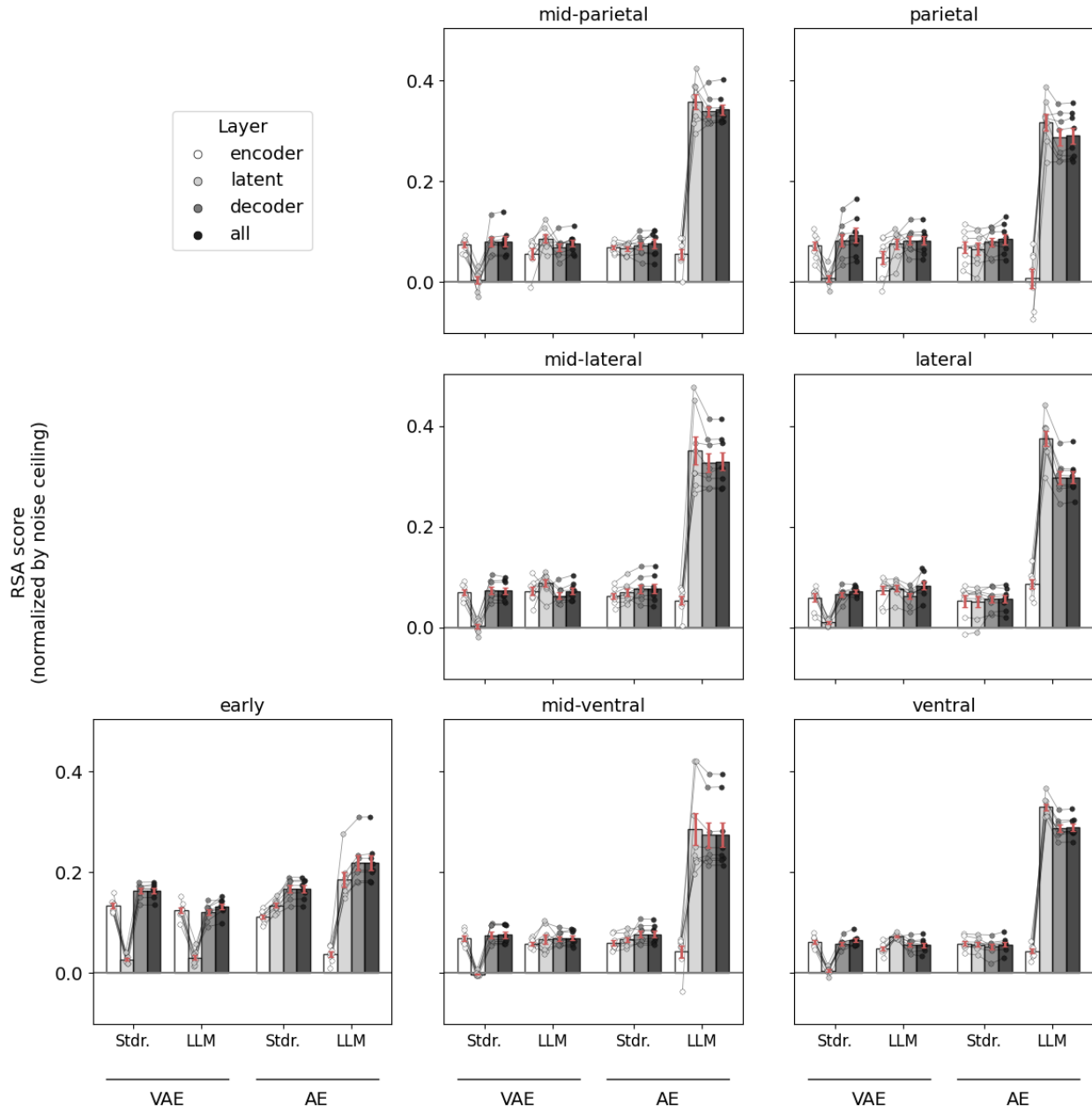


Figure 3. **(b)** RSA scores using each model component: encoder, latent variables, or decoder. The black bars show the all-combined condition (the same data as (a)).

Table 1. One-tailed paired t -tests to assess whether LLM-guided models achieved higher RSA scores than standard models. p -values were corrected for multiple comparisons to control the false discovery rate (FDR). Bold indicates FDR-corrected p -values $p_{\text{FDR}} < 0.05$. This analysis was exploratory, and the results should be interpreted descriptively.

(a) LLM-guided VAE vs. standard VAE.

ROI	Encoder		Latent		Decoder		All	
	p	p_{FDR}	p	p_{FDR}	p	p_{FDR}	p	p_{FDR}
Early	0.936	1.000	0.176	0.318	1.000	1.000	1.000	1.000
Mid-ventral	0.980	1.000	0.000	0.001	0.981	1.000	0.987	1.000
Ventral	1.000	1.000	0.000	0.000	0.791	1.000	0.987	1.000
Mid-lateral	0.337	0.591	0.000	0.001	0.997	1.000	0.673	1.000
Lateral	0.002	0.005	0.000	0.000	0.700	1.000	0.051	0.095
Mid-parietal	0.947	1.000	0.000	0.000	0.968	1.000	0.740	1.000
Parietal	0.976	1.000	0.000	0.001	0.364	0.618	0.680	1.000

(b) LLM-guided AE vs. standard AE.

ROI	Encoder		Latent		Decoder		All	
	p	p_{FDR}	p	p_{FDR}	p	p_{FDR}	p	p_{FDR}
Early	1.000	1.000	0.001	0.002	0.001	0.002	0.001	0.002
Mid-ventral	0.914	1.000	0.000	0.001	0.000	0.001	0.000	0.001
Ventral	0.975	1.000	0.000	0.000	0.000	0.000	0.000	0.000
Mid-lateral	0.833	1.000	0.000	0.001	0.000	0.000	0.000	0.000
Lateral	0.032	0.061	0.000	0.000	0.000	0.000	0.000	0.000
Mid-parietal	0.856	1.000	0.000	0.000	0.000	0.000	0.000	0.000
Parietal	0.998	1.000	0.000	0.000	0.000	0.000	0.000	0.000

The RSA scores showed a weaker effect of the LLM guidance on the VAE’s representational alignment than on the AE’s, despite both being optimized with the LLM alignment loss. This difference might stem from the VAE’s stochastic sampling, which causes its latent variables to deviate from LLM embeddings. In the deterministic AE, the LLM alignment loss targets the latent variable $\mathbf{z}_i$, whereas in the VAE it targets the posterior mean $\boldsymbol{\mu}_i$, and the latent variable $\mathbf{z}_i$ is randomly sampled from the posterior. To measure deviation from LLM embeddings, we compared cosine similarity to LLM embeddings across the VAE’s latent variables $\mathbf{z}_i$, its posterior mean $\boldsymbol{\mu}_i$, and the AE’s latent variables. Cosine similarity to LLM embeddings was lower for the VAE’s latent variables $\mathbf{z}_i$ than for the posterior mean $\boldsymbol{\mu}_i$ and for the AE’s latent variables (Figure 4). This confirmed that stochastic sampling introduced a mismatch between the VAE’s latent variables and the LLM embeddings, potentially explaining the VAE’s weaker representational alignment relative to the AE.

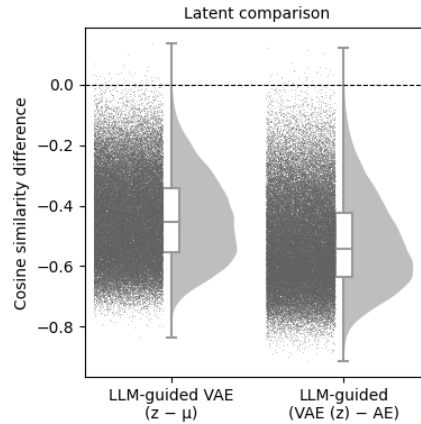


Figure 4. VAE stochasticity introduced deviation from LLM embeddings. We computed cosine similarity to LLM embedding e_i for the LLM-guided VAE’s posterior mean $\boldsymbol{\mu}_i$ and latent variables $\mathbf{z}_i$, and the LLM-guided AE’s latent variables (as in Figure 2a), then computed differences (i) $\mathbf{z}_i$ vs. $\boldsymbol{\mu}_i$ within the VAE, and (ii) VAE vs. AE latent variables.

As an additional analysis, we explored an extension of the LLM-guided VAE for text-to-image generation. Specifically, we generated images directly from LLM embeddings using the VAE decoder (Figure 4). As a reference, we trained another decoder from scratch for the same task (see Supplementary Material S2). Image generation quality from these models was poor, which could be due to design choices or a fundamental difficulty in generating images directly from LLM embeddings.



Figure 4. Example images generated from LLM embeddings using two decoder models: one transferred from the LLM-guided VAE (VAE decoder) and the other trained from scratch (decoder).

4 Discussion

In this study, we investigated whether guiding VAE’s representations toward LLM embeddings improved its representational alignment with the human visual system. We introduced an LLM alignment loss, defined as the cosine distance between the VAE posterior means for images and LLM embeddings of the corresponding image captions. The loss made the VAE’s posterior means similar to LLM embeddings while reducing image reconstruction quality (Figure 2). The LLM guidance improved representational alignment of the VAE latent variables with higher visual areas, but it did not improve the encoder, decoder, or overall alignment with the human visual system (Figure 3b). For the AE, the LLM guidance improved the representational alignment of the latent variables and decoder, but not the encoder. The AE’s overall alignment was comparable to its latent variable alignment, suggesting that the latent variables largely explain the overall representational alignment with the human visual system.

VAE stochasticity may explain its weak representational alignment. Our analysis confirmed that random sampling of latent variables from the posterior made the latent variables deviate from LLM embeddings (Figure 4). As the LLM guidance improved representational alignment for the deterministic AE (Figure 3), the deviation from LLM embeddings likely explains weak representational alignment for the LLM-guided VAE.

While LLM embeddings may help improve representational alignment, image reconstruction from them can be challenging. LLM embeddings show representational alignment with the human brain [Schrimpf et al., 2021, Caucheteux and King, 2022, Goldstein et al., 2022, Doerig et al., 2025]. Consistent with this, the LLM-guided AE, which most closely reflected LLM embeddings among our models, showed the best representational alignment (Figure 3). However, its image reconstruction quality was the worst and insufficient to capture visual experiences during mental imagery or dreaming (Figure 2).

While this trade-off might stem from our design choices, it may reflect a fundamental mismatch between one-dimensional LLM embeddings and a two-dimensional convolutional decoder. Our decoder first transforms one-dimensional LLM embeddings through a fully-connected layer, and then reshapes the resulting vector into two-dimensional spatial feature maps. This process learns a fixed mapping between LLM embedding dimensions and image

spatial locations, regardless of image content. This can be problematic because LLM embeddings encode semantic information in a distributed, non-spatial manner and may not consistently encode information relevant to a specific spatial location in the same embedding dimensions. In line with this, image generation directly from LLM embeddings yielded poor image quality (Figure 4). Several text-to-image models such as Stable Diffusion [Rombach et al., 2021] overcome this issue using a cross-attention mechanism, which bridges distributed LLM embeddings and image spatial locations [Vaswani et al., 2017]. Adapting such mechanisms while maintaining biological plausibility is an interesting direction for future work.

The LLM embedding space may also be incompatible with the VAE Gaussian prior. We obtained LLM embeddings using MPNet [Song et al., 2020], which projects texts onto a 768-dimensional unit-norm hypersphere, mapping semantically similar texts closer together. Intuitively, this embedding space appears compatible with a Gaussian prior, since high-dimensional Gaussian distributions exhibit a hyperspherical shell-like distribution, called the “soap bubble” phenomenon [Murphy, 2023]. However, some properties of the embedding space may conflict with the Gaussian prior. For example, MPNet embeddings cluster in a narrow region of the space (Supplementary Material S3), known as anisotropy [Godey et al., 2024]. Davidson et al. [2018] proposed a VAE variant with a hyperspherical latent space that may work well with the LLM embedding space. The compatibility of such geometric properties with the VAE prior should be explored further.

Approaches other than LLM guidance might better integrate linguistic information into VAEs. For example, a vision-language contrastive approach has been shown to improve representational alignment with the human visual system [Wang et al., 2023]. Incorporating such a contrastive approach into VAEs might improve representational alignment while maintaining image reconstruction quality.

Altogether, our results highlight several challenges in guiding VAEs through LLM embeddings. Addressing these challenges could not only enable text-to-image generation but also provide a computational tool for deepening our understanding of visual perception, including mental imagery and dreaming.

5 Code and Data Availability

Code is available in the GitHub repository: <https://github.com/yotaKawashima/vision-llm-vaе>. We analyzed only publicly available data: image and captions from the Microsoft Common Objects in Context (COCO) dataset [Lin et al., 2014] and fMRI data from the Natural Scenes Dataset (NSD) [Allen et al., 2022].

6 Use of AI Tools

We used Anthropic Claude Sonnet 4.6 [Anthropic, 2026] to check and improve the text and to assist with coding for the analysis.

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Supplementary Material

S1 Variational Autoencoder (VAE)

S1.1 Model Architecture

We implemented a convolutional Variational Autoencoder (VAE) (Table 1) with a 10-layer encoder and decoder. The encoder processes input images of size 128×128 pixels through ten convolutional blocks. Each block consists of a convolution layer, ReLU activation, and layer normalization. We applied MaxPooling and adaptive average pooling to progressively reduce the spatial dimensions. The output is mapped to a Gaussian posterior through two fully connected layers: one to estimate the mean μ and the other to estimate the log variance $\log \sigma^2$ (fc_mu and fc_logvar). Note that we did not model covariance terms. We set the latent dimension to 768 to match the embedding dimension of MPNet [Song et al., 2020]. We sampled latent variables $\mathbf{z}$ from the posterior using the reparameterization trick [Kingma and Welling, 2013]:

$$\mathbf{z} = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (4)$$

and fed the sampled $\mathbf{z}$ to the decoder. The decoder reconstructs input images from $\mathbf{z}$.

Supplementary Table 1. VAE Architecture. Abbreviations: s = stride, p = padding.

(a) Encoder architecture. Each convolutional block consists of Conv2d, ReLU, and LayerNorm.

Block	Spatial Size	Operation
1	128×128	7×7 conv, 32ch, s=1, p=3
2		7×7 conv, 32ch, s=1, p=3
3		5×5 conv, 64ch, s=1, p=2
2 × 2 max pooling, s=2		
4	64×64	5×5 conv, 64ch, s=1, p=2
2 × 2 max pooling, s=2		
5	32×32	3×3 conv, 128ch, s=1, p=1
6		3×3 conv, 128ch, s=1, p=1
7		3×3 conv, 256ch, s=1, p=1
8		3×3 conv, 256ch, s=1, p=1
2 × 2 max pooling, s=2		
9	16×16	1×1 conv, 512ch, s=1, p=0
2 × 2 max pooling, s=2		
10	8×8	1×1 conv, 512ch, s=1, p=0
2 × 2 adaptive average pooling		
fc_mu	fully connected layer, 2048 → 768	
fc_logvar	fully connected layer, 2048 → 768	

(b) Decoder architecture. Each convolutional block consists of Conv2d, GroupNorm, and ReLU. GroupNorm normalizes data within groups of four channels.

Block	Spatial Size	Operation
		fully connected layer, $768 \rightarrow 2048$
		bilinear upsampling
1	8×8	3×3 conv, 512ch, s=1, p=1
		bilinear upsampling
2	16×16	3×3 conv, 256ch, s=1, p=1
		bilinear upsampling
3	32×32	3×3 conv, 256ch, s=1, p=1
4		3×3 conv, 128ch, s=1, p=1
5		3×3 conv, 128ch, s=1, p=1
6		3×3 conv, 64ch, s=1, p=1
		bilinear upsampling
7	64×64	5×5 conv, 64ch, s=1, p=2
		bilinear upsampling
8	128×128	5×5 conv, 32ch, s=1, p=2
9		7×7 conv, 32ch, s=1, p=3
10		7×7 conv, 3ch, s=1, p=3

S1.2 Dataset

For model optimization, we used the Microsoft Common Objects in Context (COCO) dataset [Lin et al., 2014]. The training and validation datasets are publicly available and contain over 10k natural scene images and five captions per image. We obtained MPNet embeddings for captions, averaged them per image, and L2-normalized them. Following Doerig et al. [2025], we excluded 73,000 images from the dataset for our model optimization to avoid data overlap between the model optimization and our downstream Representational Similarity Analysis (RSA). This yielded 48,236 image–embedding pairs for training and 2,051 for validation. We used the excluded 73,000 pairs only for testing. We cropped and resized images to 128×128 pixels following the Natural Scenes Dataset (NSD) [Allen et al., 2022].

For model training, we used several data augmentation techniques. We applied random affine transformation (rotation: $\pm 2^\circ$, height and width: $\pm 5\%$), color jittering (brightness, contrast, and saturation: $\pm 10\%$), random grayscaling (probability: 5%), and Gaussian blurring (kernel size: 3×3 pixel, σ : randomly drawn from $[0.1, 2.0]$). We normalized images using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). No data augmentation was applied for validation and testing.

S1.3 Training Procedure

We trained the model in three stages to direct the VAE latent posteriors toward MPNet embeddings. Throughout all stages, we used the AdamW optimizer [Loshchilov and Hutter, 2019] with a learning rate of 5×10^{-4} , weight decay of 1×10^{-6} , and gradient clipping with a maximum norm of 1.0. We also used cosine annealing as a learning rate scheduler [Loshchilov and Hutter, 2017] across 50 epochs with a batch size of 128 and the minimum learning rate of 1×10^{-6} . We stored model weights every 5 epochs and stopped training once validation performance plateaued. Note that all hyperparameters were selected based on validation performance.

Stage 1: Encoder pre-training. We pre-trained the encoder to predict MPNet embeddings $\mathbf{e}_i$ for input images $\mathbf{x}_i$. The mean estimation layer (fc_mu) predicted the embeddings $\mathbf{e}_i$. We excluded the log variance estimation layer (fc_logvar) at this stage. We optimized the encoder using a loss $\mathcal{L}_{\text{stage1}}$:

$$\mathcal{L}_{\text{stage1}} = \alpha \mathcal{L}_{\text{align}} + \mathcal{L}_{\text{norm}} \quad (5)$$

where α controls the weight of the alignment loss ($\alpha = 10$), $\mathcal{L}_{\text{align}}$ is the cosine distance between the posterior mean $\boldsymbol{\mu}_i$ and the LLM embedding $\mathbf{e}_i$, and $\mathcal{L}_{\text{norm}}$ encourages the posterior mean to have unit norm to match the L2-normalized MPNet embeddings. Given N pairs of images and LLM embeddings, we can express $\mathcal{L}_{\text{align}}$ as follows:

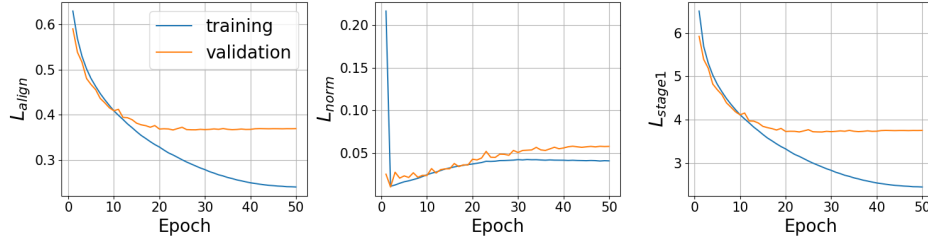
$$\mathcal{L}_{\text{align}} = \frac{1}{N} \sum_{i=1}^N \left(1 - \frac{\boldsymbol{\mu}_i \cdot \mathbf{e}_i}{\|\boldsymbol{\mu}_i\| \|\mathbf{e}_i\|} \right) \quad (6)$$

$\mathcal{L}_{\text{norm}}$ is defined using the Smooth L1 loss ℓ_{smoothL1} :

$$\ell_{\text{smoothL1}}(x, y) = \begin{cases} \frac{(x-y)^2}{2} & \text{if } |x-y| < 1 \\ |x-y| - \frac{1}{2} & \text{otherwise} \end{cases} \quad (7)$$

$$\mathcal{L}_{\text{norm}} = \frac{1}{N} \sum_{i=1}^N \ell_{\text{smoothL1}}(\|\boldsymbol{\mu}_i\|, 1) \quad (8)$$

Figure 1 shows these Stage 1 losses during training and validation. Based on the validation performance, we selected model weights from epoch 30 for the following stages.

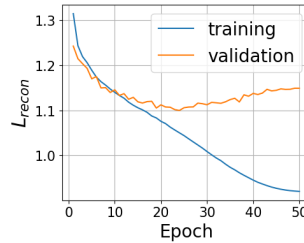


Supplementary Figure 1. Stage 1 training and validation loss. From left to right: LLM alignment loss $\mathcal{L}_{\text{align}}$, Smooth L1 loss $\mathcal{L}_{\text{norm}}$, and total loss $\mathcal{L}_{\text{stage1}} = \alpha \mathcal{L}_{\text{align}} + \mathcal{L}_{\text{norm}}$ with $\alpha = 10$.

Stage 2: Decoder pre-training. We pre-trained the decoder to reconstruct images from the LLM embeddings predicted by the pre-trained encoder. We kept the encoder frozen during this stage. The decoder receives the output from the mean estimation layer (fc_mu). We optimized the decoder using a reconstruction loss $\mathcal{L}_{\text{recon}}$:

$$\mathcal{L}_{\text{recon}} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 \quad (9)$$

where $\hat{\mathbf{x}}_i$ is the reconstructed image given the input image $\mathbf{x}_i$. Based on validation performance, we selected model weights from epoch 25 for the next stage (Figure 2).

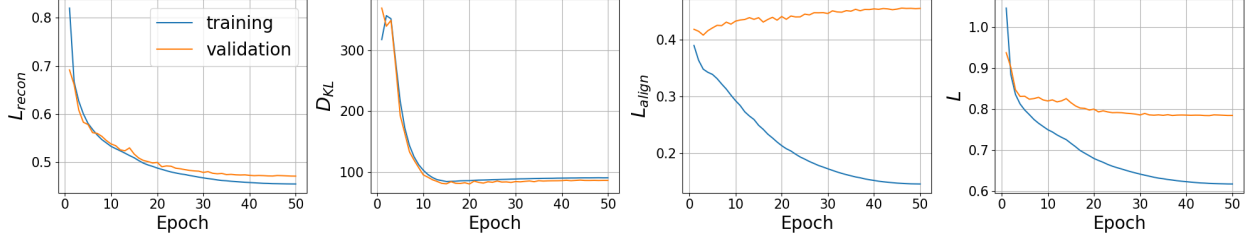


Supplementary Figure 2. Stage 2 training and validation image reconstruction loss $\mathcal{L}_{\text{recon}}$.

Stage 3: Full VAE training. We trained the full VAE using the pre-trained encoder and decoder weights for initialization. At this stage, we optimized the variance estimation layer (fc_logvar) for the first time. We used a combined loss $\mathcal{L}$:

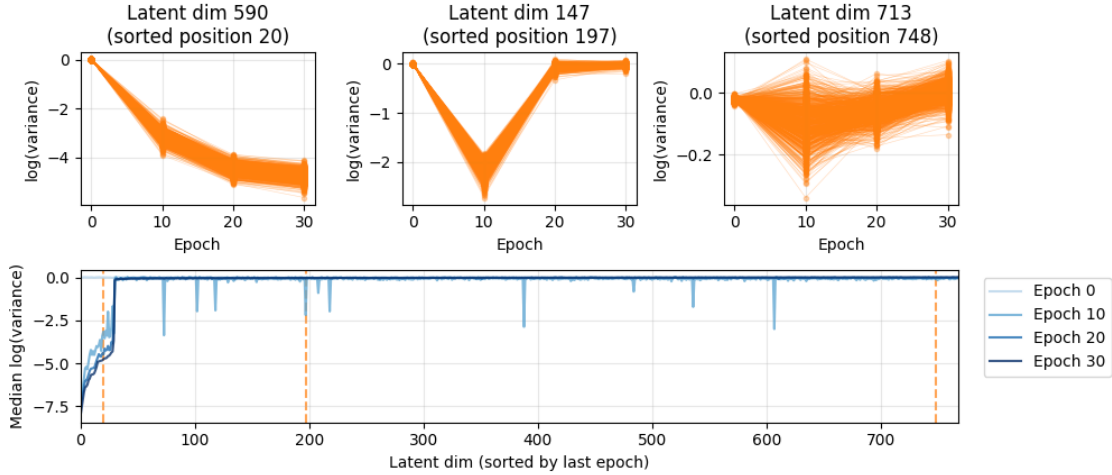
$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta D_{\text{KL}}(q(\mathbf{z}|\mathbf{x})\|p(\mathbf{z})) + \gamma \mathcal{L}_{\text{align}} \quad (10)$$

where D_{KL} is the Kullback-Leibler (KL) divergence between the approximate posterior $q(\mathbf{z}|\mathbf{x})$ over latent variable $\mathbf{z}$ and the standard Gaussian prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$, β is the weight of the KL term, and γ is the weight of the LLM alignment loss ($\gamma = 0.5$). To avoid a common VAE training failure, known as posterior collapse [Bowman et al., 2015, Chen et al., 2016, van den Oord et al., 2017, Yang et al., 2017], we annealed β from 0 to 0.001 over the first third of the training epochs using a cosine scheduler. Based on validation performance, we selected the model at epoch 30 as the fully-trained LLM-guided VAE (Figure 3).



Supplementary Figure 3. Stage 3 training and validation loss. From left to right: image reconstruction loss $\mathcal{L}_{\text{recon}}$, Kullback-Leibler divergence between posterior and prior D_{KL} , LLM alignment loss $\mathcal{L}_{\text{align}}$, and total loss $\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta D_{\text{KL}} + \gamma \mathcal{L}_{\text{align}}$ with $\gamma = 0.5$ and β annealing from 0 to 0.001.

Since the variance estimation layer (fc_logvar) was optimized for the first time at this stage, we tracked the VAE posterior variance σ_i^2 over training. Stochasticity decreased for some posterior dimensions but remained stable for most (Figure 4).

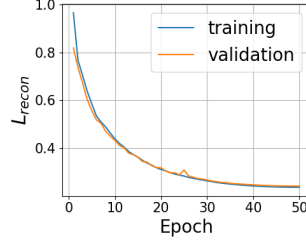


Supplementary Figure 4. VAE posterior variance σ_i^2 over training. Top three panels: variance of three example latent dimensions for 1,000 images randomly selected from the test dataset of 73,000 images. Bottom panel: median variance across the test dataset for each latent dimension, sorted by variance at the last epoch. Orange dotted lines indicate the three example dimensions.

S1.4 Reference models

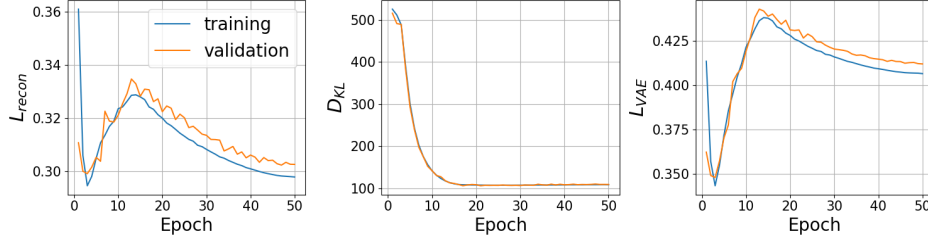
We considered three reference models: a standard AE, an LLM-guided AE, and a standard VAE.

Our AEs shared the same architecture as the LLM-guided VAE except for the log variance estimation layer (fc_logvar) and the reparameterization trick. We trained the standard AE using only the reconstruction loss $\mathcal{L}_{\text{recon}}$ (Eq. 9) and selected the model at epoch 50 as the fully-trained model (Figure 5). As the fully-trained LLM-guided AE, we used the model from Stage 2 of the LLM-guided VAE training.



Supplementary Figure 5. Standard Autoencoder (AE) training and validation loss (image reconstruction loss $\mathcal{L}_{\text{recon}}$).

Our standard VAE shared the architecture with the LLM-guided VAE but not the training procedure. We initialized the model with weights from the fully-trained standard AE and trained it without the LLM alignment loss $\mathcal{L}_{\text{align}}$ (Eq. 10). We selected the model at epoch 50 as the fully-trained standard VAE (Figure 6).



Supplementary Figure 6. Standard VAE training and validation loss. From left to right: image reconstruction loss $\mathcal{L}_{\text{recon}}$, Kullback-Leibler divergence between posterior and prior D_{KL} , and total loss $\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \beta D_{\text{KL}}$ with β annealing from 0 to 0.001.

S2 Text-to-image generation

We explored an extension of the LLM-guided VAE for text-to-image generation. The VAE decoder reconstructs images from latent variables, which are trained to align with LLM embeddings. Therefore, it should also be able to generate images from LLM embeddings. However, LLM embeddings have unit norm while latent variables do not. To address this difference, we introduced a learnable scaling layer that receives an LLM embedding e_i and outputs a scalar scaling factor. The rescaled embedding e'_i is passed to the frozen VAE decoder for image generation. We also introduced a loss $\mathcal{L}_{\text{norm-vae}}$ that encourages e'_i and its corresponding VAE posterior mean μ_i to have a similar norm. Using ℓ_{smoothL1} from Eq. 7, $\mathcal{L}_{\text{norm-vae}}$ is defined as:

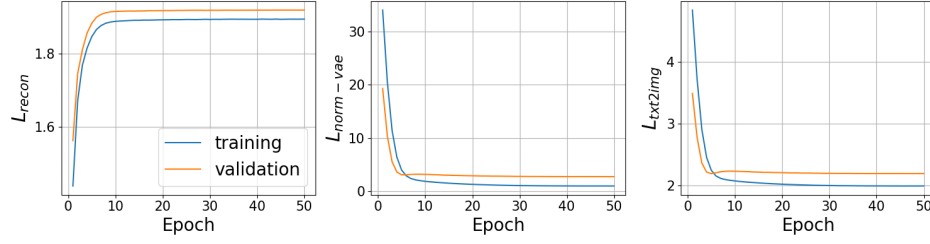
$$\mathcal{L}_{\text{norm-vae}} = \frac{1}{N} \sum_{i=1}^N \ell_{\text{smoothL1}}(\|\mu_i\|, \|e'_i\|) \quad (11)$$

We trained the scaling layer with a total loss $\mathcal{L}_{\text{txt2img}}$:

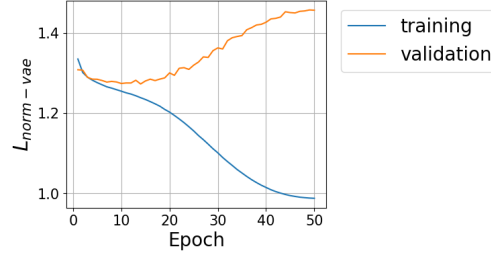
$$\mathcal{L}_{\text{txt2img}} = \delta \mathcal{L}_{\text{norm-vae}} + \mathcal{L}_{\text{recon}} \quad (12)$$

where δ controls the weight of $\mathcal{L}_{\text{norm-vae}}$ ($\delta = 0.1$), and $\mathcal{L}_{\text{recon}}$ is the image reconstruction loss between the generated and corresponding original images (Eq. 9). We also trained a decoder from scratch to reconstruct images directly from LLM embeddings using $\mathcal{L}_{\text{recon}}$ only. Figure 7 shows training and validation loss from these models. Together with Figure 4 in the main text, these results suggest that training was challenging. This could be due to our design choices, such as insufficient training data or suboptimal hyperparameters. Alternatively, generating images from LLM embeddings might simply be fundamentally difficult.

(a) VAE decoder



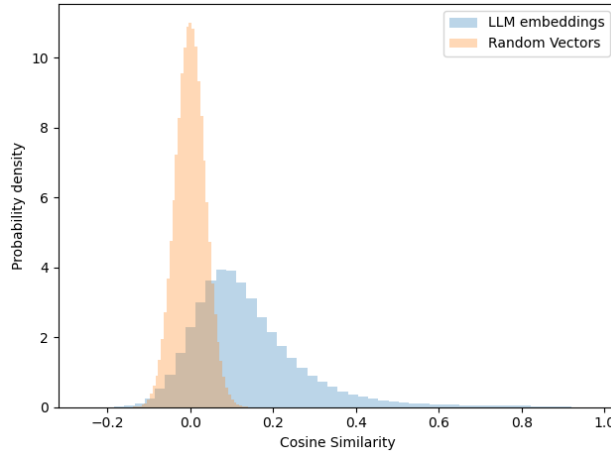
(b) Decoder



Supplementary Figure 7. Training and validation loss for text-to-image generation models. We used two decoder models: one transferred from the LLM-guided VAE (VAE decoder) and the other trained from scratch (decoder). Training and validation loss for (a) VAE decoder and (b) decoder. Based on the validation loss, we selected the VAE decoder at epoch 50 and the decoder at epoch 10 as the fully trained models.

S3 MPNet embedding anisotropy

We examined geometric properties of MPNet embeddings [Song et al., 2020]. We computed pairwise cosine similarities between MPNet embeddings for 1,000 samples randomly selected from the test dataset. We compared the distribution to that of 1,000 random vectors drawn from the standard normal distribution. Figure 8 suggests that MPNet embeddings cluster in a narrow region of the space, a property known as anisotropy [Godey et al., 2024].



Supplementary Figure 8. Distributions of pairwise cosine similarities computed from 1,000 MPNet embeddings (blue) and 1,000 random vectors drawn from a standard normal distribution (orange).